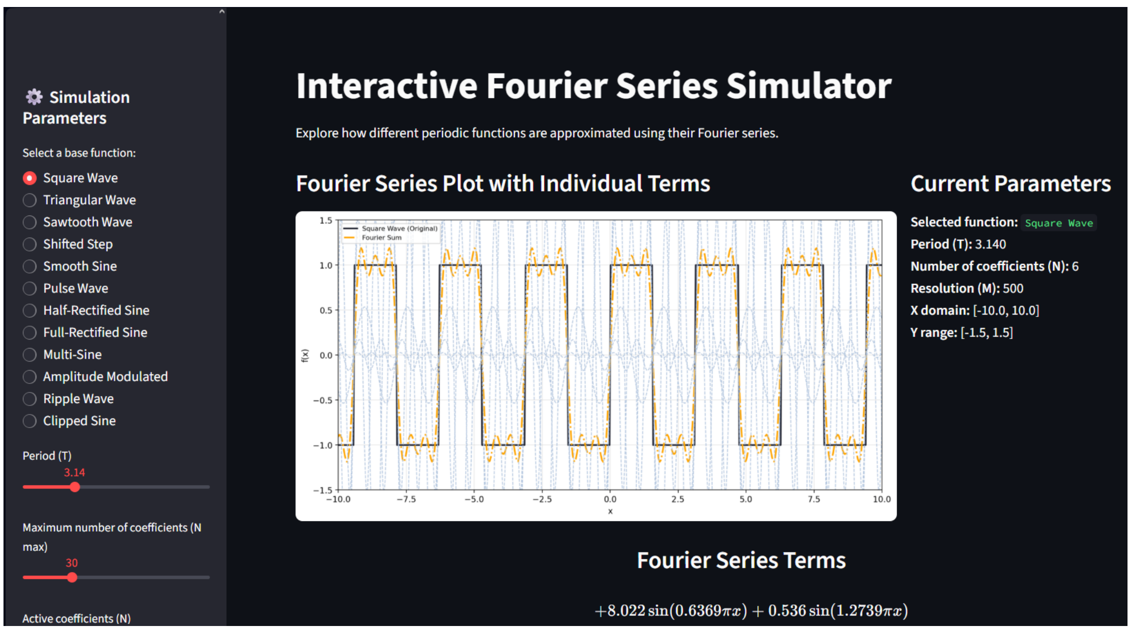
Interactive Fourier Series Simulator

User Manual



Juan Ángel Gamez Díaz

Mathematical Methods for Physicists — Project

Introduction

The *Interactive Fourier Series Simulator* is a web application designed to help students and researchers visualize the Fourier decomposition of periodic functions.

Through an intuitive and dynamic interface, users can:

- Select a base periodic function.
- Adjust physical and numeric parameters such as the period T, resolution, and number of Fourier coefficients.
- Visualize the original function, the Fourier approximation, and each individual sine/cosine term.
- Obtain a clean mathematical representation of the Fourier series in LaTeX format.

This tool is especially useful for courses in mathematical methods, signals, physics, and general Fourier analysis.

System Requirements

- A modern browser (Chrome, Edge, Firefox, Opera).
- Internet connection (if hosted online) or Python environment with:
  - Python 3.10 or newer
  - Streamlit
  - NumPy
  - Matplotlib
  - SQLite (for optional suggestions database)

Home Interface Overview

Upon opening the simulator, the user will find:

- A main plot area displaying the Fourier approximation.
- A sidebar menu used to control simulation parameters.

- A mathematical panel showing the Fourier series expression.
- An optional suggestion box for requesting new periodic functions.

The design follows a clean and minimalistic style, with soft colors and emphasized headings.

Sidebar Parameters

The sidebar control panel allows the user to adjust:

Base Function Selection

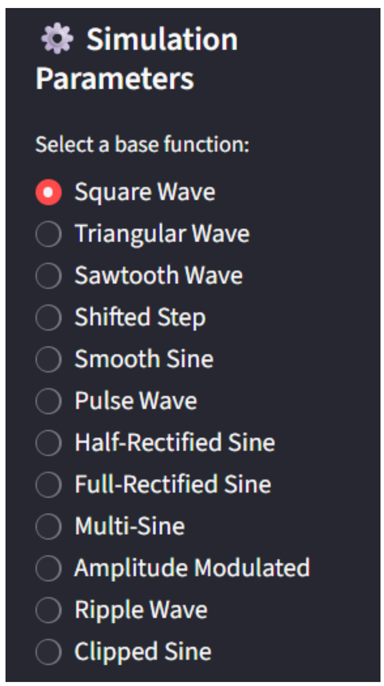


Figure 1: Selection of the function.

Choose a periodic function from the internal catalog (`waves_dic`), such as:

- Square wave
- Triangle wave
- Sawtooth
- Custom sinusoidal forms

Period T

Controls the fundamental periodicity of the function:

$$f(x+T) = f(x).$$

There exists a function in the list that does not satisfy this definition.

Maximum Number of Coefficients $N_{\max}$

Sets the upper limit for available Fourier terms.

Active Coefficients N

Defines how many terms are actually used in the Fourier reconstruction:

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N} \left(a_n \cos(\frac{2\pi n}{T}x) + b_n \sin(\frac{2\pi n}{T}x)\right)$$

Sampling Resolution M

Number of points used to discretize the domain.

Higher M gives smoother curves.

Plot Ranges

- $x_{\min}, x_{\max}$
- $y_{\min}, y_{\max}$

These allow zooming in/out of specific regions.

Fourier Series Terms Panel

A special block of the interface displays the analytical Fourier series using LaTeX:

$$+8.022 \sin(0.6369\pi x)$$

$$+0.536 \sin(1.2739\pi x)$$

$$+2.958 \sin(1.9108\pi x)$$

Two-column presentation is used for cleaner reading.
All coefficients are rounded to three significant digits to reduce noise.

Plot Area

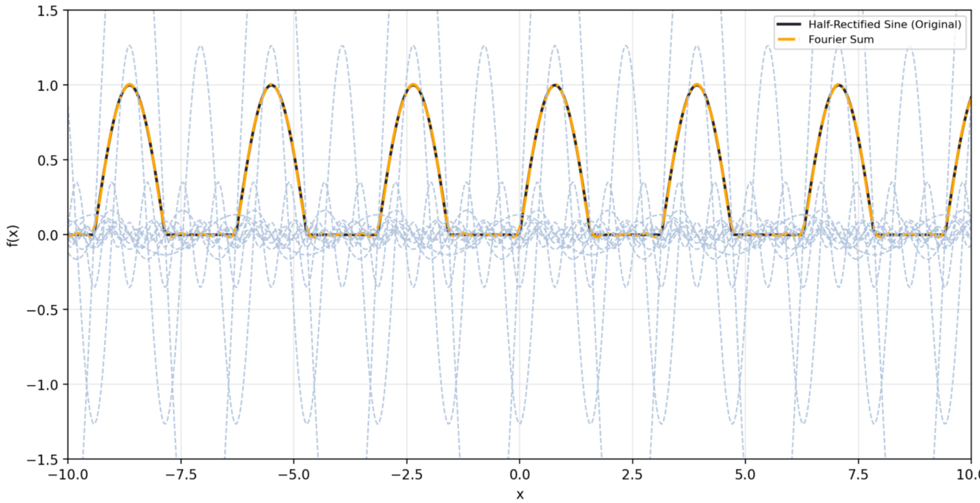


Figure 2: Plot Area.

The main graph shows:

- Original function (in dark blue)
- Fourier reconstructed sum (in orange)
- Individual Fourier terms (in light blue dashed lines)

These visualization choices make it easy to understand the contribution of each harmonic.

Suggestion Box

At the bottom of the app, users can submit requests for additional periodic functions. Suggestions are saved in a local SQLite database with timestamps.

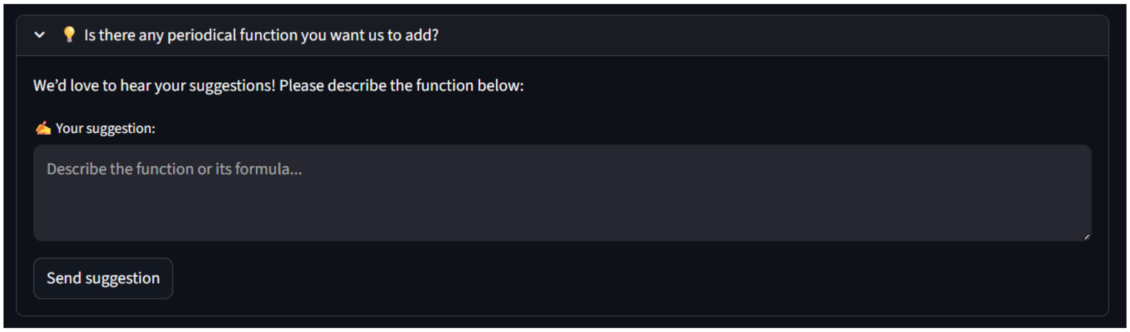


Figure 3: Suggestion Box

Example Workflow

1. Select *Square Wave*.
2. Set $T = 3.14$.
3. Choose $N = 6$.
4. Set resolution to $M = 500$.
5. Observe how the Fourier approximation converges.
6. Check the LaTeX term visualization to analyze harmonic dominance.

Credits

This simulator was created by:

Juan Ángel Gamez Díaz
Physics Engineering — Computer Vision Researcher

as part of the *Mathematical Methods for Physicists* course project.